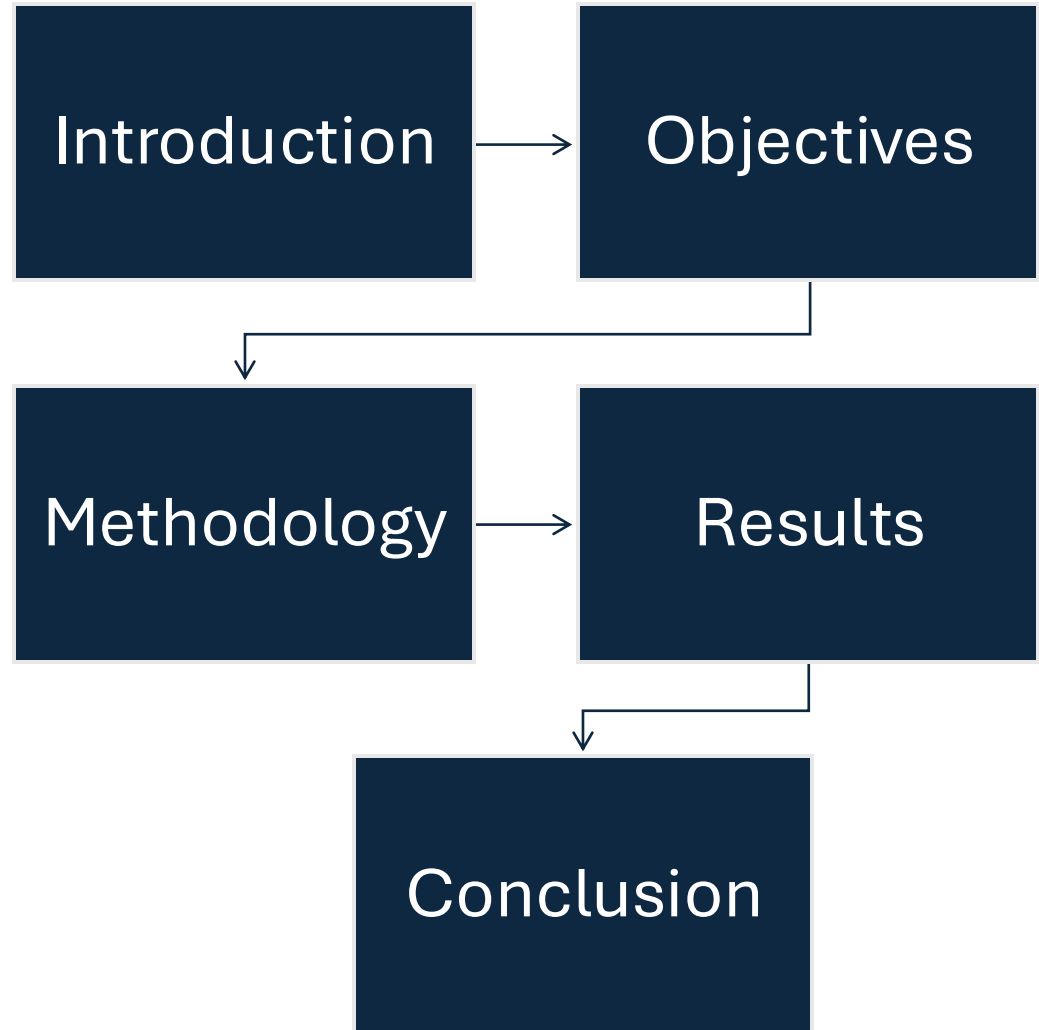




Microarray-Based Expression Profiling for Identification of Differentially Expressed Genes and Related Pathways in Breast Cancer

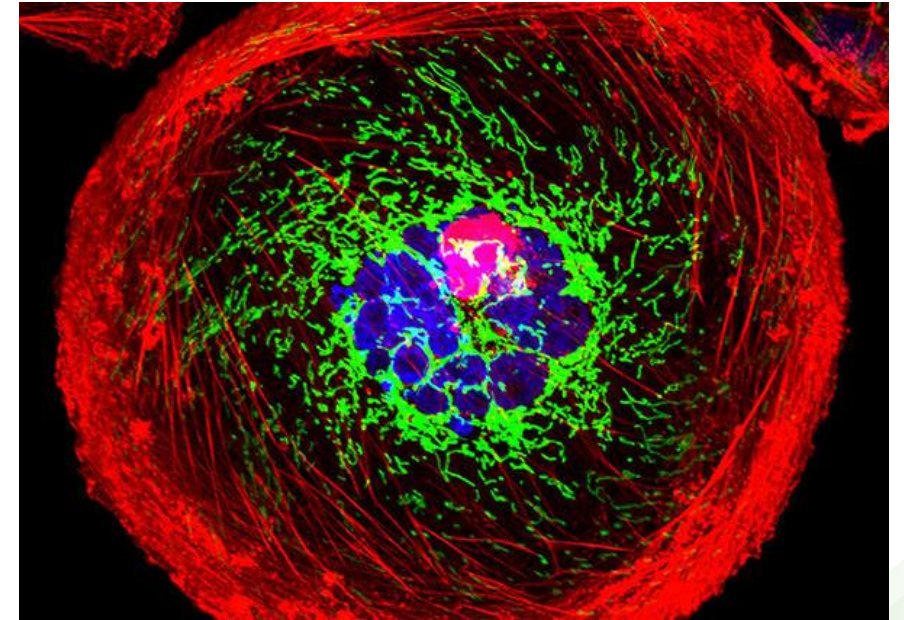
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- Supervised by: Saddia Bano
- Department: Department of Biological Sciences
- Virtual University of Pakistan

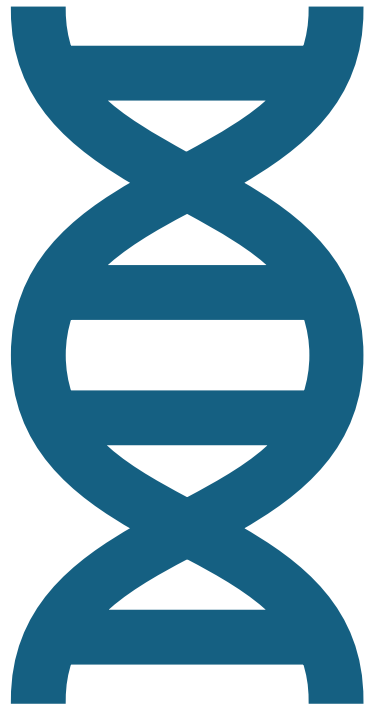
Outline



Introduction (Breast Cancer)

- Breast cancer is one of the most common types of cancer in women, making up about 1 in 4 cancer cases
- The breast is not a vital organ for staying alive, and early-stage breast cancer is not life-threatening.
- Cancer cells are abnormal, they don't stick together well and can break off easily.
- These cells can then travel through the blood or lymph system and spread to other parts of the body (this is called metastasis).
- Even though diagnosis and treatment have improved, many patients still face poor outcomes and survival rates.





Why Gene Expression Studies?

- Old methods can only look at a few genes at a time, so they don't give the full picture.
- Gene expression studies (like microarrays) let scientists study thousands of genes at once.
- This helps us better understand how cancer works at the genetic level.
- It's useful for:
 - Finding early signs (biomarkers) of cancer
 - Discovering new treatment targets
 - Understanding why some cancers grow fast or resist treatment
 - Overall, it helps doctors give more accurate and personalized treatments.

Objectives of the study

1

- Identification of differentially expressed genes



2

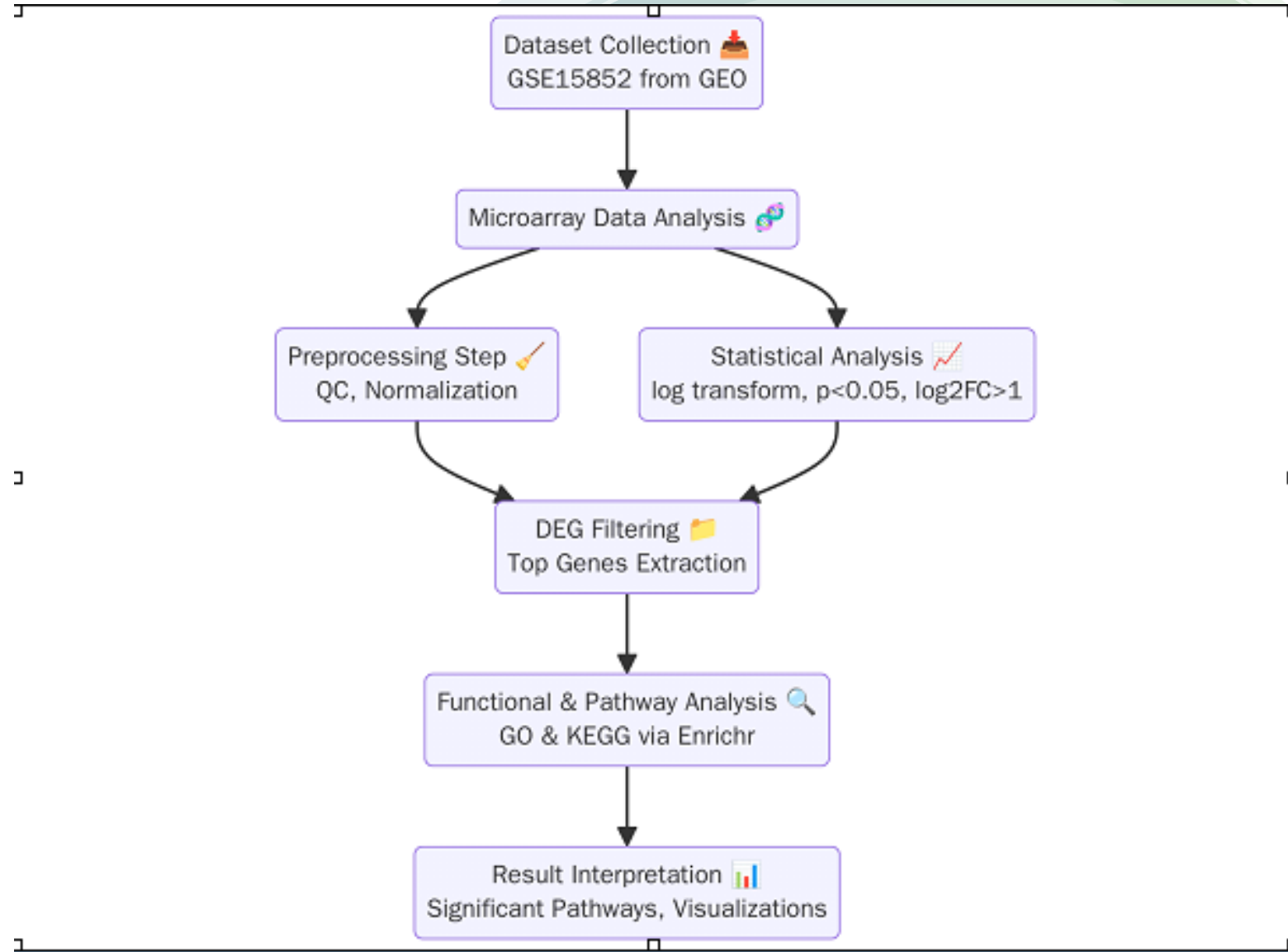
- Identification of related metabolic pathways of breast cancer



Dataset Information

- **GSE15852** (43 breast tumor, 43 normal)
- Source: **GEO database**
- Platform: **Affymetrix Human Genome U133A Array**

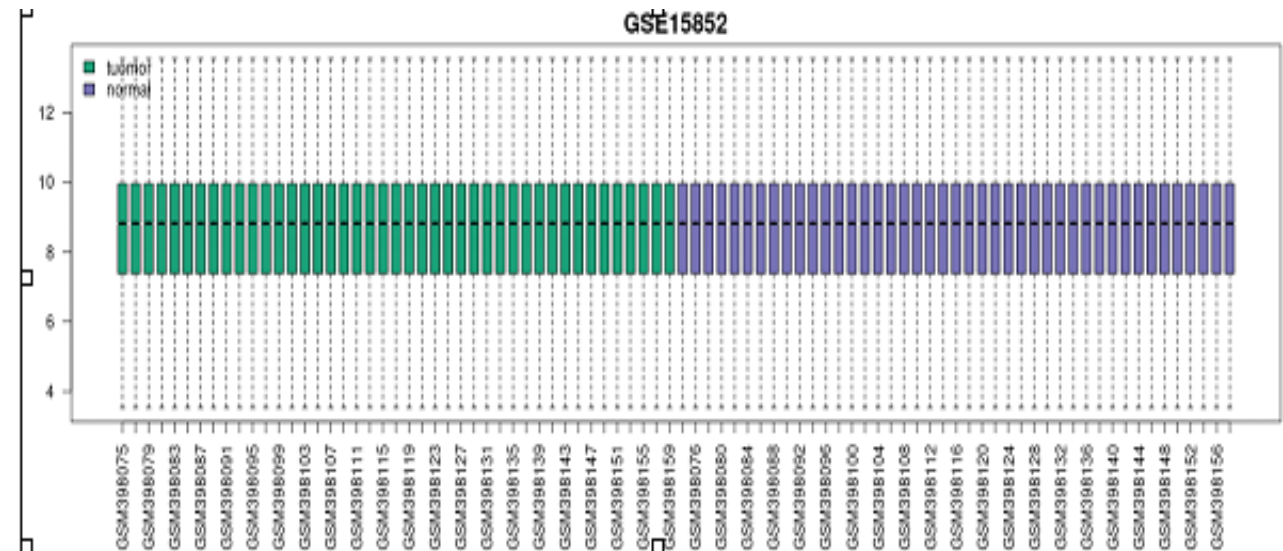
Workflow Diagram



Results

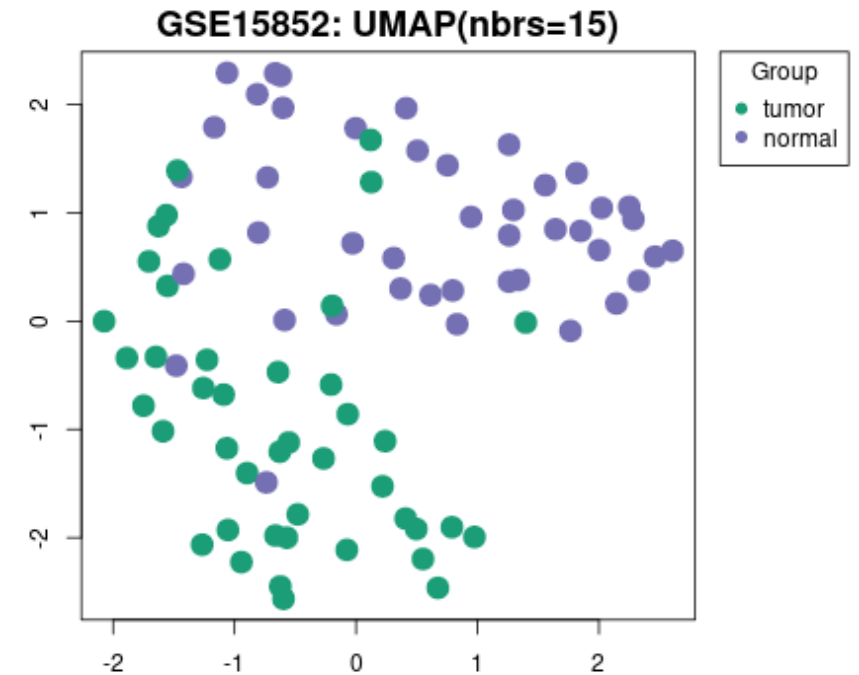
Box Plot (QC)

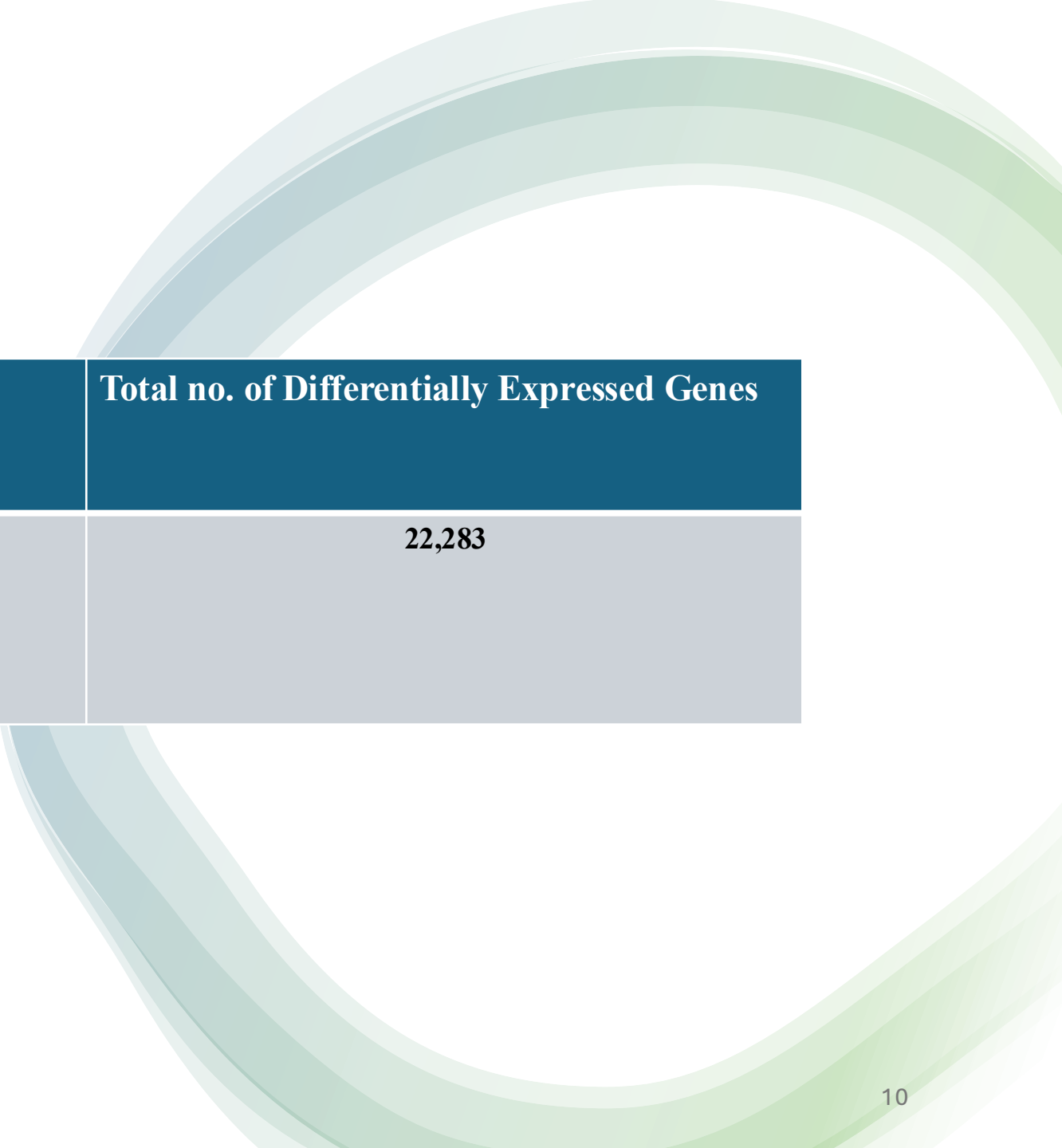
- Confirms successful normalization of samples
- Comparable distribution across tumor and normal groups



UMAP Plot

- “This UMAP shows that tumors cluster separately from normals, which confirms that our dataset can distinguish disease vs healthy samples.”

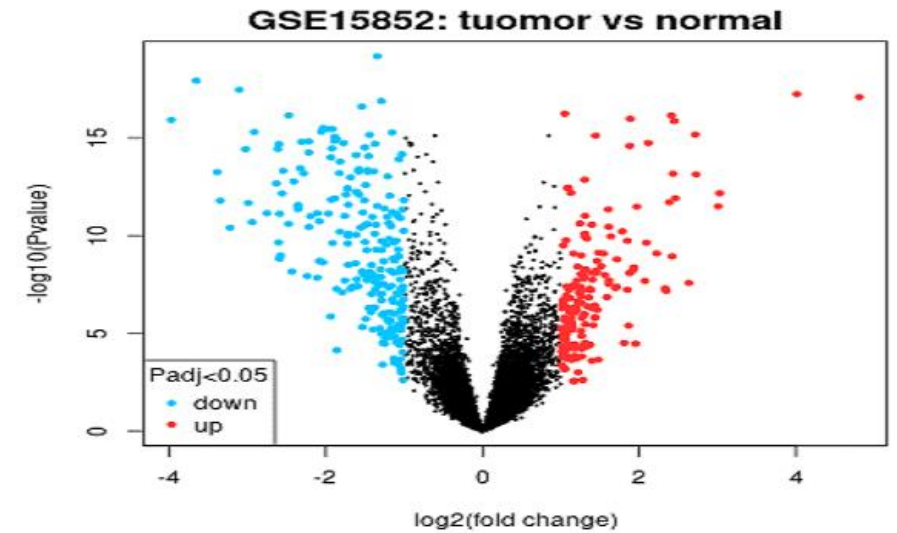




Dataset	Threshold	Total no. of Differentially Expressed Genes
GSE15852	p value < 0.05 FC < -1 FC > 1	22,283

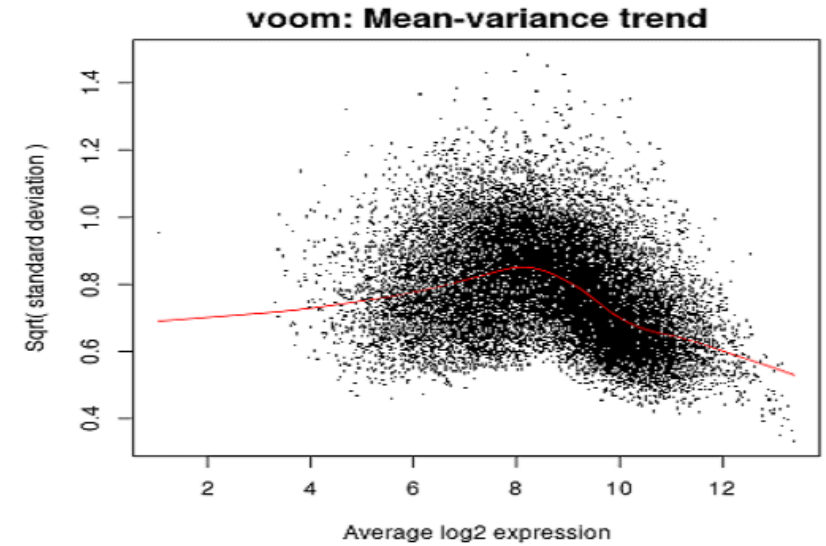
Volcano Plot

- Visualizes significant DEGs
- Red = upregulated, Blue = downregulated



Voom Plot

- Displays mean–variance relationship
- Confirms variance modelling suitability

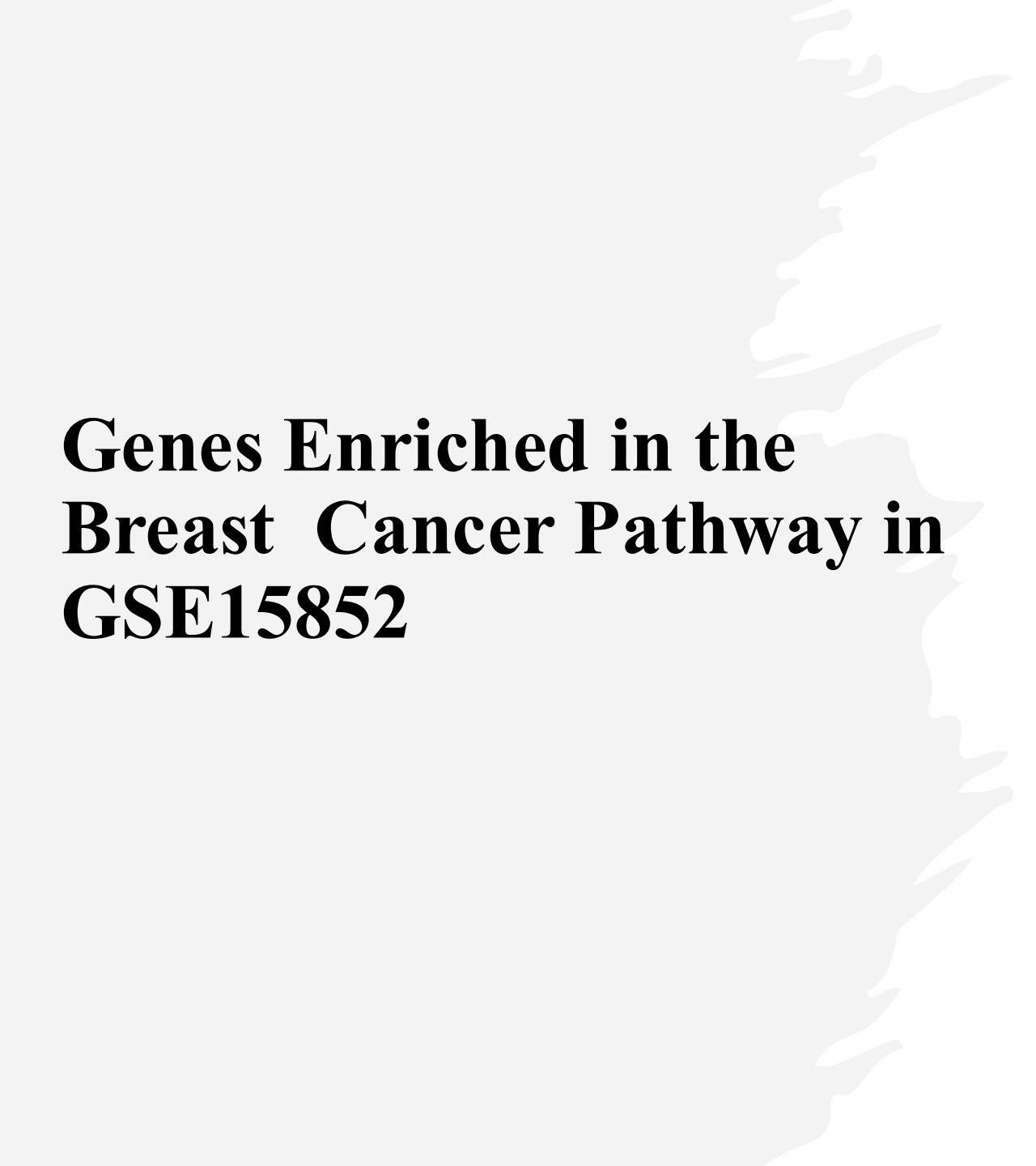


Pathway Enrichment Analysis

Term	No of genes
Pathways in cancer	85
Breast Cancer	26
Proteoglycans in cancer	49

Genes Enriched in the Breast Cancer Pathway in GSE15852

Genes	Log2FC
RB1	-0.226344
FLT4	0.090087
LEF1	0.423752
PTEN	0.105720
FGF2	-0.040633
EGFR	-0.071073
IGF1R	0.076756
FGF5	-0.192238
ERBB2	-0.248987
E2F3	-0.069058
HES1	0.198759
FZD1	-0.084788



Genes Enriched in the Breast Cancer Pathway in GSE15852

GENE	LOG2FC
JUN	0.168994
FZD5	-0.074398
FZD4	0.629876
CSNK1A1	-0.059070
FZD7	-0.071162
WNT5A	0.017050
IGF1	-0.030359
FOS	0.114189
ESR1	0.244603
CDK4	0.098931
FGF18	-0.089761
CTNNB1	-0.103093
SOS1	-0.029171
FGFR1	-0.147828



Discussion

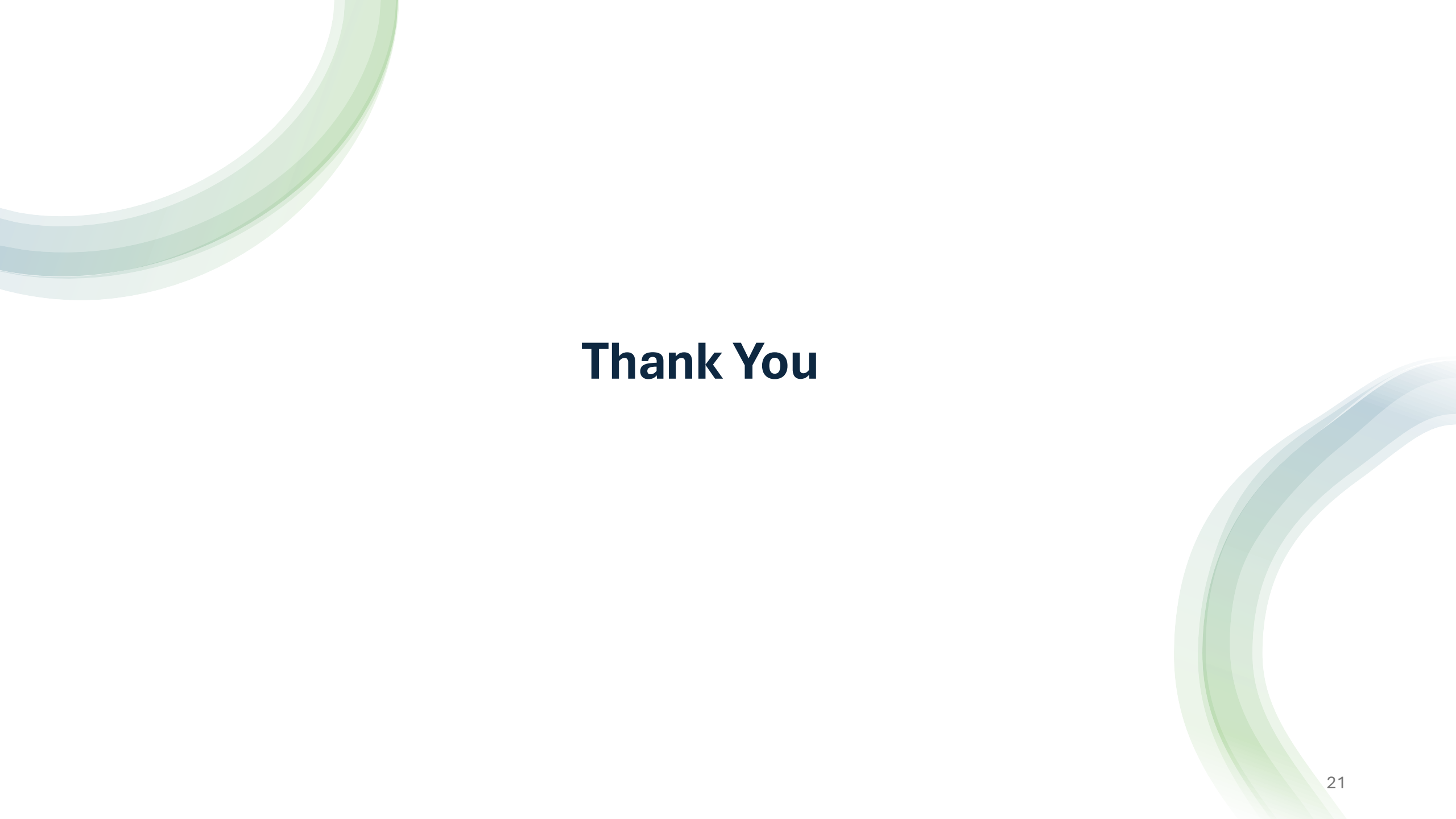
Gene	Description
CDK4	CDK4 is a cell cycle regulator. Its upregulation promotes uncontrolled cell division by allowing cells to pass through the G1/S checkpoint, contributing to tumor growth. This explains why CDK4/6 inhibitors are effective in breast cancer treatment.
FGF18	FGF18 is part of the fibroblast growth factor family that normally drives cell growth and angiogenesis. Here it is slightly downregulated, suggesting that these cancer samples may not depend heavily on FGF18 for progression.
CTNNB1	CTNNB1 encodes β -catenin, a key player in the Wnt pathway. Its downregulation may indicate reduced β -catenin activity, but since other Wnt genes are up, the pathway overall still appears active. This balance influences tumor invasion and survival.
SOS1	SOS1 links receptor tyrosine kinases to the RAS/MAPK pathway. Its slight downregulation suggests weaker activation of this growth pathway in these samples.
FGFR1	FGFR1 is a receptor for fibroblast growth factors. Normally, its amplification drives aggressive tumors, but its downregulation here suggests it is not a major driver in these breast cancer samples.
FZD5	FZD5 is a receptor in the Wnt pathway. Its downregulation suggests it plays a smaller role in Wnt signaling here, compared to other Frizzled receptors like FZD4.
FZD4	FZD4 is strongly upregulated, showing very active Wnt signaling. This promotes stemness, tumor progression, and therapy resistance, making it a key gene in this dataset.
CSNK1A1	CSNK1A1 is a kinase that regulates β -catenin stability in Wnt signaling. Its slight downregulation may alter Wnt signaling balance, influencing cancer cell survival.
FZD7	FZD7 is another Wnt receptor. Its downregulation suggests it is not the dominant Wnt activator here, unlike FZD4 which is strongly up.

Gene	Description
WNT5A	WNT5A is a Wnt ligand. Its slight upregulation suggests some contribution to non-canonical Wnt signaling, which supports invasion and metastasis.
IGF1	IGF1 normally stimulates growth through IGF1R. Its downregulation suggests reduced ligand-driven signaling, though IGF1R itself is slightly upregulated.
FOS	FOS is part of the AP-1 transcription factor. Its upregulation supports estrogen signaling and drives cancer cell growth and invasion.
ESR1	ESR1 encodes the estrogen receptor alpha (ER α). Its upregulation confirms that these breast cancers are estrogen-dependent, typical of luminal subtypes.
ERBB2	ERBB2 (HER2) is normally amplified in HER2+ cancers. Its downregulation here indicates these cancers are not HER2-driven, possibly luminal or basal-like subtypes.
E2F3	E2F3 controls cell cycle gene expression. Its downregulation may reduce E2F-driven proliferation, but other cell cycle pathways are active, such as CDK4 upregulation.
HES1	HES1 is a transcription factor in the Notch pathway. Its upregulation helps maintain cancer stem-like properties and therapy resistance.
FZD1	FZD1 is a Wnt receptor. Its downregulation indicates it is not strongly contributing to Wnt signaling in these samples.
JUN	JUN, along with FOS, forms the AP-1 complex. Its upregulation drives transcription of genes involved in proliferation and survival.

Gene	Description
LEF1	LEF1 is a Wnt transcription factor. Its strong upregulation shows active Wnt/ β -catenin signaling, which supports tumor growth and invasion.
PTEN	PTEN is a tumor suppressor that inhibits the PI3K/AKT pathway. Surprisingly, it is slightly upregulated here, which may represent a subtype where PTEN is retained.
FGF2	FGF2 usually promotes angiogenesis and tumor growth. Its slight downregulation suggests weaker contribution to cancer progression in this dataset.
EGFR	EGFR is important in triple-negative breast cancer. Its downregulation here suggests these samples are not strongly dependent on EGFR signalling.
IGF1R	IGF1R is a receptor that promotes survival and endocrine resistance. Its upregulation here shows that IGF signaling remains active despite lower IGF1 levels.
FGF5	FGF5 promotes angiogenesis and growth. Its downregulation suggests it is not playing a strong role in these breast cancers.
RB1	RB1 is a tumor suppressor controlling the cell cycle. Its downregulation removes cell cycle brakes, promoting uncontrolled tumor cell division.
FLT4	FLT4 (VEGFR3) drives lymphangiogenesis. Its upregulation indicates higher potential for lymph node spread in breast cancer.

Conclusion

- The project successfully identified key gene expression changes and associated biological pathways in breast cancer.
- These findings contribute to the growing body of molecular knowledge on breast cancer and provide potential leads for future biomarker discovery.



Thank You